SLIDING MODE CONTROL

CONSIDER A SIMO SYSTEM (ROBOT JOINTS SEPARATELY).

$$\dot{\underline{x}} = A\underline{x} + \underline{b}u + f(x)$$

UNMODELLED DYNAMICS

WE DEFINE THE SLIDING MODE CONTROL ACTION AS

$$u = u_{eq} + u_{sw}$$

EQUIVALENT CONTROLLER

$$u_{eq} = -\underline{k}^T \underline{x}$$

SWITCHING CONTROLLER IS OBTAINED FROM THE SLIDING SURFACE.

$$\sigma(\underline{x}) = \underline{h}^T(\underline{x} - \underline{x}_d)$$

h IS THE RIGHT EIGENVECTOR FOR THE CLOSED LOOP SYSTEM. TAKE DERIVATIVE OF σ

$$\begin{aligned}
\dot{\sigma}(x) &= \underline{h}^T(\dot{\underline{x}} - \dot{\underline{x}}_d) \\
&= \underline{h}^T(A\underline{x} + \underline{b}u + F(x) - \dot{x}_d) \\
&= \underline{h}^T(A\underline{x} + \underline{b}u_{eq} + \underline{b}u_{sw} + F(x) - \dot{x}_d) \\
&= \underline{h}^T(A\underline{x} - \underline{b}\underline{k}^T\underline{x} + \underline{b}u_{sw} + F(x) - \dot{x}_d)
\end{aligned}$$

SINCE $A_c = A - \underline{b}\underline{k}^T$ IS OUR CLOSED LOOP SYSTEM MATRIX

$$\dot{\sigma}(\underline{x}) = \underline{h}^T(A_c\underline{x} + \underline{b}u_{sw} + F(\underline{x}) - \dot{\underline{x}}_d)$$

SINCE $\underline{h}^T A_c = 0$

$$\dot{\sigma}(\underline{x}) = \underline{h}^T\left(\underline{b}\, u_{sw} + F(x) - \dot{\underline{x}}_d\right).$$

REARRANGING GIVES.

$$u_{sw} = (\underline{h}^T\underline{b})^{-1}\left(\underline{h}^T\dot{\underline{x}}_d - \underline{h}^T F(\underline{x}) + \dot{\sigma}(\underline{x})\right)$$

WE DEFINE $\dot{\sigma}(\underline{x}) = \underline{h}^T \Delta f - \eta\, \mathrm{sgn}(\sigma(\underline{x}))$.

WHERE $\Delta F = f(\underline{x}) - \hat{f}(\underline{x})$.

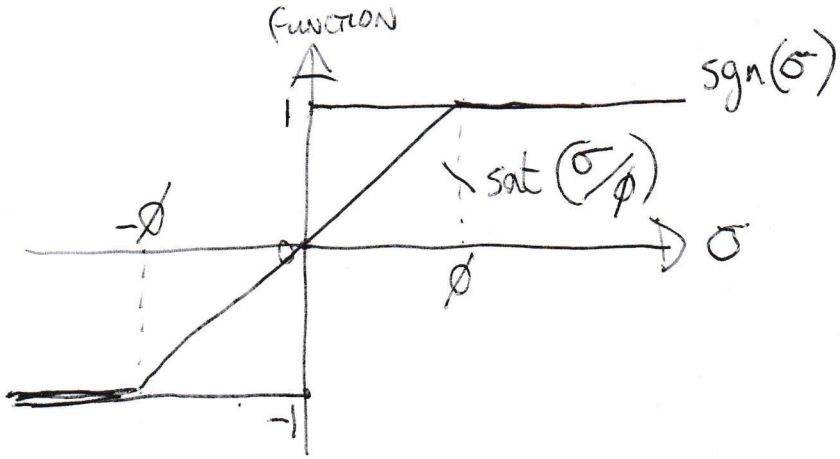
$$\therefore\ u_{sw} = (\underline{h}^T\underline{b})^{-1}\left(\underline{h}^T\dot{\underline{x}}_d - h^T\left(f(\underline{x}) - \Delta F\right) - \eta\, \mathrm{sgn}(\sigma(x))\right)$$

$$u_{sw} = (\underline{h}^T\underline{b})^{-1}\left(\underline{h}^T\dot{\underline{x}}_d - h^T\hat{f}(\underline{x}) - \eta\, \mathrm{sgn}(\sigma(x))\right)$$

CONTROLLER

$$u = -\underline{k}^T\underline{x} + (\underline{h}^T\underline{b})^{-1}\left(\underline{h}^T\dot{\underline{x}}_d - \underline{h}^T\hat{f}(\underline{x}) - \eta\, \mathrm{sgn}(\sigma(\underline{x}))\right)$$

CHATTERING.



HARD SWITCHING.

NO ESTIMATE OF UNMODELLED DYNAMICS $(\hat{f}(\underline{x}) = 0)$

$$u = -\underline{k}^T\underline{x} + (\underline{h}^T\underline{b})^{-1}\left(\underline{h}^T\dot{\underline{x}}_d - \eta\, \mathrm{sgn}(\sigma)\right)$$

SOFT SWITCHING $\quad u = -\underline{k}^T\underline{x} + (\underline{h}^T\underline{b})^{-1}\left(\underline{h}^T\dot{\underline{x}}_d - \eta\, \mathrm{sat}(\sigma/\phi)\right).$